Unbiased Estimators & Confidence Intervals

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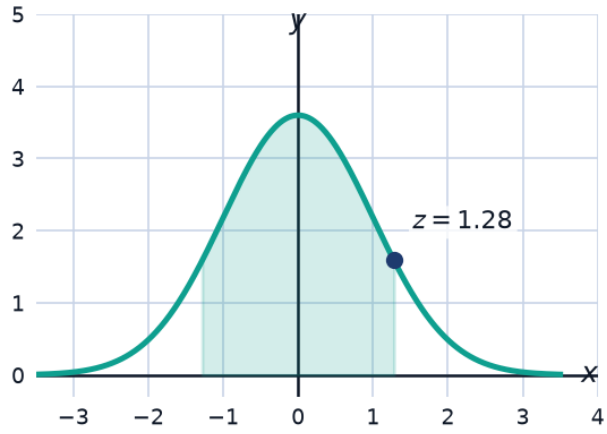
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A sample of size $n = 49$ has mean $\bar{x} = 44$, drawn from a population with $\mu_0 = 47$ and known $\sigma = 10$. Find the test statistic z, correct to 2 decimal places.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $X \sim N(\mu, \sigma^2)$, approximately what percentage of values lie within 3 standard deviations of the mean? Show your complete algebraic work.



3 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

If $X \sim N(\mu, \sigma^2)$, approximately what percentage of values lie within 1 standard deviation of the mean? Show your complete algebraic work.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The masses, x grams, of 50 packets of tea gave $\sum x = 5090$ and $\sum x^2 = 518380$. (a) Calculate unbiased estimates of the population mean and variance. (b) Calculate a 95% confidence interval for the population mean.

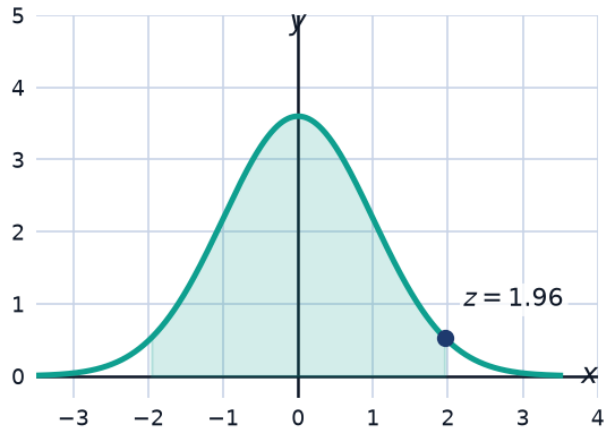
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A continuous dataset is grouped into class intervals. The interval $20 \le x < 35$ has a frequency of 45. Find the frequency density for this class.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A sample of size 100 from a population (known $\sigma = 20$) gives $\bar{x} = 142$. Find a 99% CI for μ and state what "99% confidence" means.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the points $A(1, 2, 3)$, $B(2, 4, 1)$, and $C(-1, 0, 5)$ in $\mathbb{R}^3$. Find a non-zero vector $\mathbf{n}$ normal to the plane containing A, B, C, and write the Cartesian equation of the plane.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · Advanced Find the extreme values of $f(x, y) = x^2 + 2y^2$ on the circle constraint $g(x, y) = x^2 + y^2 = 1$ using the Method of Lagrange Multipliers.

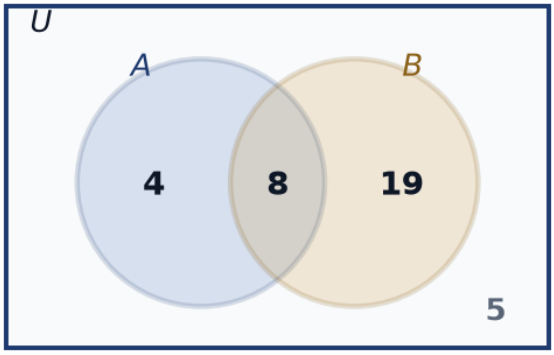
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 3 · AdvancedEvaluate the definite integral analytically: $\int_{-3}^{3} \left(x^5\sqrt{9 - x^2} + \sqrt{9 - x^2}\right) dx$.

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Find the binomial expansion of $\frac{1}{\sqrt{4-x}}$ in ascending powers of x up to and including the term in x^2. State the set of values of x for which the expansion is valid.



11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 3 · Advanced A tank initially contains 100 liters of pure water. Brine containing 0.5 kg of salt per liter enters the tank at a rate of 4 L/min, and the well-stirred solution leaves the tank at the same rate of 4 L/min. Find the amount of salt $S(t)$ in the tank after t minutes.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Level 3 · Advanced Evaluate the improper integral or establish its divergence: $\int_0^\infty xe^{-x/2}\,dx$

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Level 3 · Advanced Find the full interval of convergence (including endpoint behavior) for the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n^2 \cdot 4^n}$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Use the substitution $u = 2x + 1$ to find the exact value of $\int_0^4 \frac{x}{\sqrt{2x+1}}\,dx$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

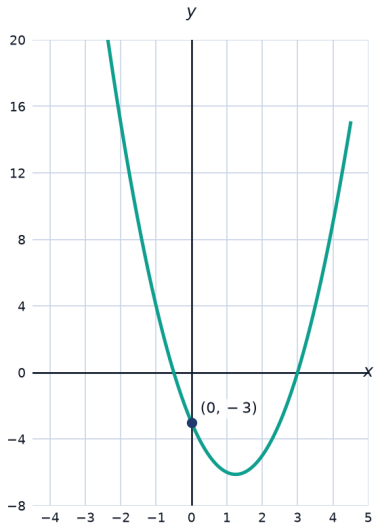
15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 3.01 3D Vectors & the Cross Product**

A random variable X has distribution $\text{Po}(\lambda)$. A single observation of X is taken to test $H_0 : \lambda = 6$ against $H_1 : \lambda < 6$ at the 5% level. (a) Find the critical region. (b) State the actual significance level. (c) Find the probability of a Type II error if the true value of λ is 3.

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16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions**

Solve the quadratic equation $2x^2 - 5x - 3 = 0$ by completing the square. State the vertex of the corresponding parabola $y = 2x^2 - 5x - 3$.



17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)**

Prove by mathematical induction that for all integers $n \geq 1$: $1 + 3 + 5 + \ \ldots\ + (2n - 1) = n^2$.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.02 The Poisson Distribution**

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2 + 1}{x^2 - 4}$. Describe its end behavior using limits.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Further Integration Techniques**

Level 3 · Advanced A particle moves along a curve defined parametrically by $x = t^2 + 1$ and $y = t^3 - 3t$. Find the value of $\frac{d^2y}{dx^2}$ at $t = 2$ and determine whether the curve is concave up or concave down at this point.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.02 Differential Equations**

The circle C has equation $x^2 + y^2 - 6x + 4y - 12 = 0$. (a) Find the coordinates of the centre and the exact radius of C. (b) Find the equation of the tangent to C at the point $P(6, 2)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given sample observations $x_1, \ldots, x_n$, what is the formula for the unbiased estimator s_{n-1}^2 of the population variance σ^2?

a. $s_{n-1}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$

b. $s_n^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$

c. $s^2 = \frac{1}{n+1}\sum_{i=1}^{n}x_i^2$

d. $s^2 = \frac{(\sum x_i)^2}{n-1}$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A sample of $n = 5$ values has $\sum x_i = 30$ and $\sum x_i^2 = 210$. Calculate the unbiased estimate of the population variance s_{n-1}^2.

a. 7.5

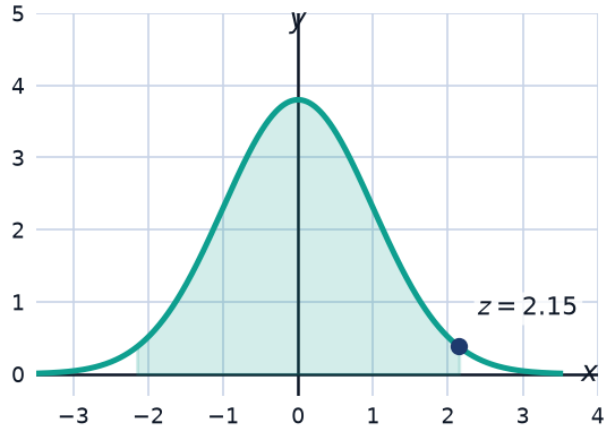
b. 6.0

c. 30

d. 9.0

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If a sample standard deviation computed with denominator n is $s_n = 4.0$ for $n = 16$, what is the unbiased standard deviation s_{n-1}?



a. $s_{n-1} = 4.0 \times \sqrt{\frac{16}{15}} \approx 4.13$

b. $s_{n-1} = 4.0 \times \sqrt{\frac{15}{16}} \approx 3.87$

c. $s_{n-1} = 4.0$

d. $s_{n-1} = 4.0 \times \frac{16}{15} \approx 4.27$

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A random sample of $n = 64$ items from a population with known $\sigma = 12$ has mean $\bar{x} = 50$. What is the 95% confidence interval for μ ($z^* = 1.96$)?

a. $[47.06, 52.94]$

b. $[48.50, 51.50]$

c. $[44.12, 55.88]$

d. $[49.63, 50.37]$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

To reduce the margin of error of a confidence interval by a factor of 3, the sample size must be multiplied by:

a. 9

b. 3

c. $\sqrt{3}$

d. 6

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the critical value z^* for a 99% confidence interval?

a. 2.576

b. 1.960

c. 1.645

d. 2.326

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

When constructing a confidence interval for μ with unknown population variance σ^2 from a normal sample of size $n = 10$, which distribution is used?

a. Student's t-distribution with 9 degrees of freedom

b. Standard Normal Z-distribution

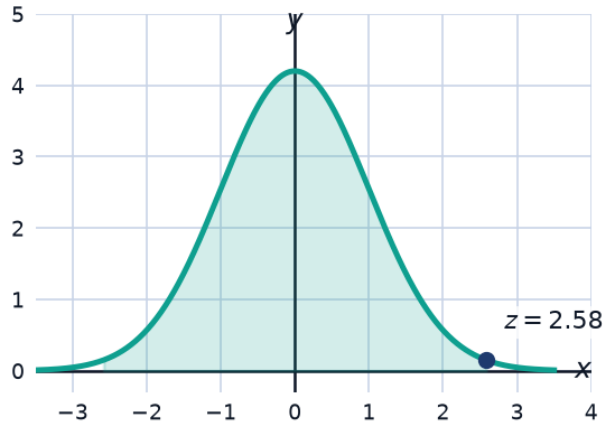
c. Student's t-distribution with 10 degrees of freedom

d. Chi-square distribution with 9 degrees of freedom

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How does the critical value t^* compare to z^* for the same confidence level and finite sample size?



a. $t^* > z^*$, because the t-distribution has heavier tails to reflect estimation uncertainty

b. $t^* < z^*$, because t is more compact

c. $t^* = z^*$ always

d. t^* can be smaller or larger depending on $\bar{x}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A sample of $n = 16$ has $\bar{x} = 22.0$ and $s_{n-1} = 2.0$. Given $t_{0.025,\,15} = 2.131$, find the 95% confidence interval for μ.

a. $[20.93, 23.07]$

b. $[21.47, 22.53]$

c. $[19.87, 24.13]$

d. $[20.00, 24.00]$

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which statement correctly conveys the meaning of a '95% confidence level'?

a. If many random samples are drawn and intervals constructed, 95% of them will contain the true population parameter

b. There is a 95% probability that the true parameter lies inside the specific interval $[47, 53]$

c. 95% of sample data values lie between 47 and 53

d. The sample mean has a 95% chance of being correct

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If a 95% confidence interval for μ is $[102, 118]$, what is the conclusion for the two-tailed test $H_0 : \mu = 100$ vs $H_1 : \mu \neq 100$ at $\alpha = 0.05$?

a. Reject H_0 at the 5% level, because 100 lies outside the 95% confidence interval

b. Fail to reject H_0, because 100 is close to 102

c. Accept H_0 as proven

d. The test is inconclusive without a P-value

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What happens to the width of a confidence interval if the confidence level is increased from 90% to 99% (with sample size fixed)?

a. The interval widens

b. The interval narrows

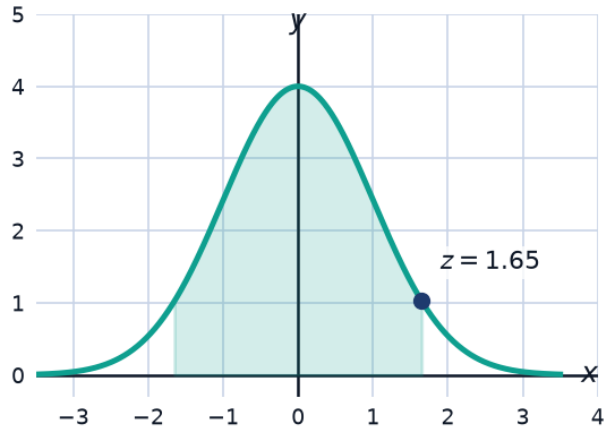
c. The interval width remains identical

d. The interval shifts to the right

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Fourier series period 2π: $a_0 =$



a. $\frac{1}{\pi}\int_{-\pi}^{\pi} f(x),dx$

b. $\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x),dx$

c. $\int_{-\pi}^{\pi} f(x)\cos x\,dx$

d. $\frac{2}{\pi}\int_{-\pi}^{\pi} f(x)\cos x\,dx$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A transition matrix is $T = \left(\frac{4}{5}, \frac{3}{5}; \frac{1}{5}, \frac{2}{5}\right)$, rows/cols ordered A, B. Find $P(B|B)$, the probability of moving from state B to state B.

a. $\frac{1}{10}$

b. $\frac{2}{5}$

c. 0

d. $\frac{1}{5}$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A transition matrix is $T = \left(\frac{4}{5}, \frac{2}{5}; \frac{1}{5}, \frac{3}{5}\right)$, rows/cols ordered A, B. Find $P(A|A)$, the probability of moving from state A to state A.

a. $\frac{4}{5}$

b. $\frac{7}{10}$

c. $\frac{3}{5}$

d. $\frac{2}{5}$

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A transition matrix is $T = \left(\frac{2}{5}, \frac{3}{5}; \frac{3}{5}, \frac{2}{5}\right)$, rows/cols ordered A, B. Find $P(B|A)$, the probability of moving from state A to state B.

a. 1

b. $\frac{4}{5}$

c. $\frac{1}{5}$

d. $\frac{3}{5}$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A transition matrix is $T = \left(\frac{4}{5}, \frac{4}{5}; \frac{1}{5}, \frac{1}{5}\right)$, rows/cols ordered A, B. Find $P(A|B)$, the probability of moving from state B to state A.

a. 1

b. $\frac{2}{5}$

c. $\frac{3}{5}$

d. $\frac{4}{5}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.02 The Poisson Distribution

Describe the locus $|z - (4 - 2i)| = 4$ on the Argand diagram.

a. $\text{Circle, centre } (4, -2), \text{ radius } 16$

b. $\text{Circle, centre } (-4, 2), \text{ radius } 4$

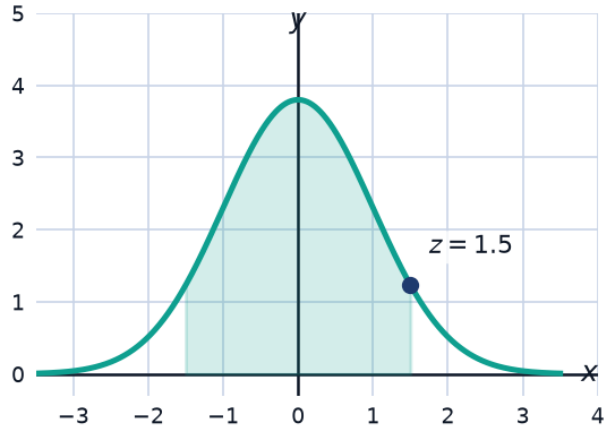
c. $\text{Line through } (4, -2)$

d. $\text{Circle, centre } (4, -2), \text{ radius } 4$

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Further Integration Techniques

A 2-state Markov chain has transition matrix $T = \left(\frac{9}{10}, \frac{1}{10}; \frac{3}{10}, \frac{7}{10}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.



a. $\frac{3}{5}$

b. $\frac{2}{5}$

c. $\frac{1}{2}$

d. $\frac{3}{4}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.02 Differential Equations

A system starts in state A. The transition matrix (rows/cols A, B) is $T = \left(\frac{4}{5}, \frac{1}{5}; \frac{3}{5}, \frac{2}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.

a. $\frac{3}{5}$

b. $\frac{7}{10}$

c. $\frac{19}{25}$

d. $\frac{2}{5}$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Find the value of k for which $(4, 3; 4, k)$ is singular (has no inverse).

a. 1

b. -3

c. 3

d. 5

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

X takes values $0, 1, 2$ with probabilities $\frac{1}{5}, \frac{3}{10}, \frac{1}{2}$. Find $\mathrm{Var}(X)$.

a. $\frac{61}{100}$

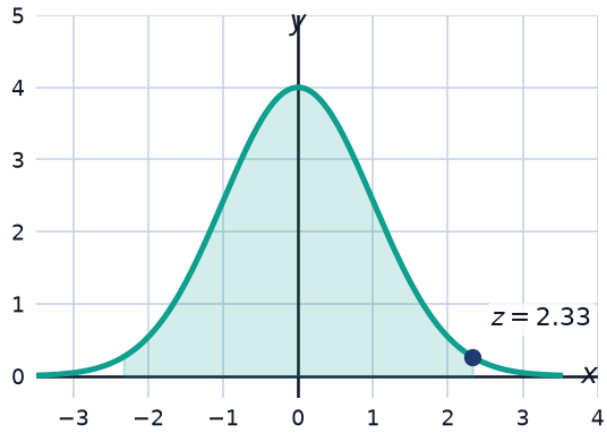
b. $\frac{121}{100}$

c. $\frac{73}{100}$

d. $\frac{4}{5}$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Matrices (AI HL)

If $X \sim B(150, \frac{1}{100})$, use a Poisson approximation to find $P(X = 1)$, correct to 4 decimal places.



a. 0.2231

b. 0.1255

c. 0.2510

d. 0.3347

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Given $\sin A = \frac{8}{17}$, $\cos A = \frac{15}{17}$, $\sin B = \frac{3}{5}$, $\cos B = \frac{4}{5}$ (both acute), find $\sin(A + B)$.

a. $\frac{69}{85}$

b. $\frac{82}{85}$

c. $\frac{77}{85}$

d. $\frac{16}{17}$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.03 Markov Chains (AI HL)

In a survey of 150 people, 74 are category A and 74 have trait X, with 12 being both A and X. Find $P(A|X)$.

a. $\frac{3}{5}$

b. $\frac{137}{150}$

c. $\frac{6}{37}$

d. $\frac{9}{37}$

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